Systematic Equities Case Packet

Click here for historical data and sample code!

1. Case overview

The world

The Archipelago has four islands:

- Cascadia Prime, in the north
- Coral Key and Azure Isle, in the east
- Palm Atoll, in the south-west

Ten companies and two ETFs trade on the Archipelago Exchange. The companies are in three groups:

- **Island businesses (5 companies)** own sites on the islands: a port, resorts, fish farms and fleets, the power grid and ferry terminals.
- **Rebuilders (3 companies)** repair the damage: a construction company, a materials supplier and a generator rental company.
- **Insurers (2 companies)** pay for the insured losses: a primary insurer and its reinsurer.

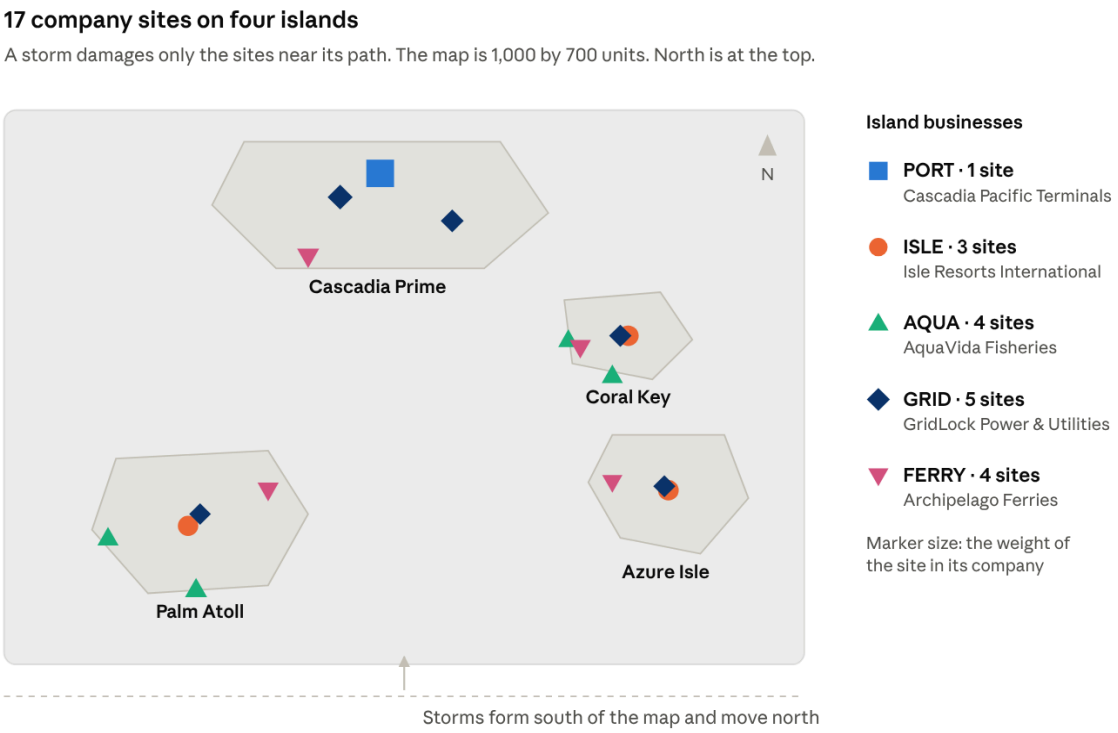


Figure 1. The 17 company sites. Marker shape shows the company; marker size shows how much of the company's value the site carries. Coordinates and weights are in `case.json`.

The season

The case is one hurricane season with three rounds. In each round, one hurricane forms south of the islands and moves north. A hurricane has these effects:

- It damages the sites near its path. The damage is permanent.
- It closes a site while the wind at that site is too strong.
- It gives work to the rebuilders.
- It causes insured losses that the insurers must pay.

Damage accumulates over the season: each storm adds to what the earlier storms did.

Nobody knows the path of a new storm in advance. Forecasters publish outlooks, rumors circulate on the wire, and prices react to both, not always correctly.

Your task

Your team trades the 10 stocks and the 2 ETFs on the Archipelago Exchange. You trade against the house and the other teams. You can trade by hand, with bots, or both. Most of the trading should be done algorithmically since there are lots of symbols and data!

At the end of round 3, each symbol settles at its fundamental value. That is the company's value after all of the season's damage. Your task is to find these values before the market does, and to trade on them.

2. How the case runs

The case has 3 rounds. Each round is one market day with one storm. Three things carry over from one round to the next:

- your positions
- the damage to each company
- the total insured losses of the season

Round	Market day	Events
Round 1	Day 1	Storm 1.
Round 2	Day 2	Storm 2. Each company starts from its condition after storm 1.
Round 3	Day 3	Storm 3. At the close, each symbol settles at its fundamental value.

Announcements can change the economics of a company at any point in the season.

The clock

The exchange uses a fast market clock. One market day, from 09:30 to 16:00, takes 13 real minutes. One real second is 30 market seconds. All feed timestamps use the market clock.

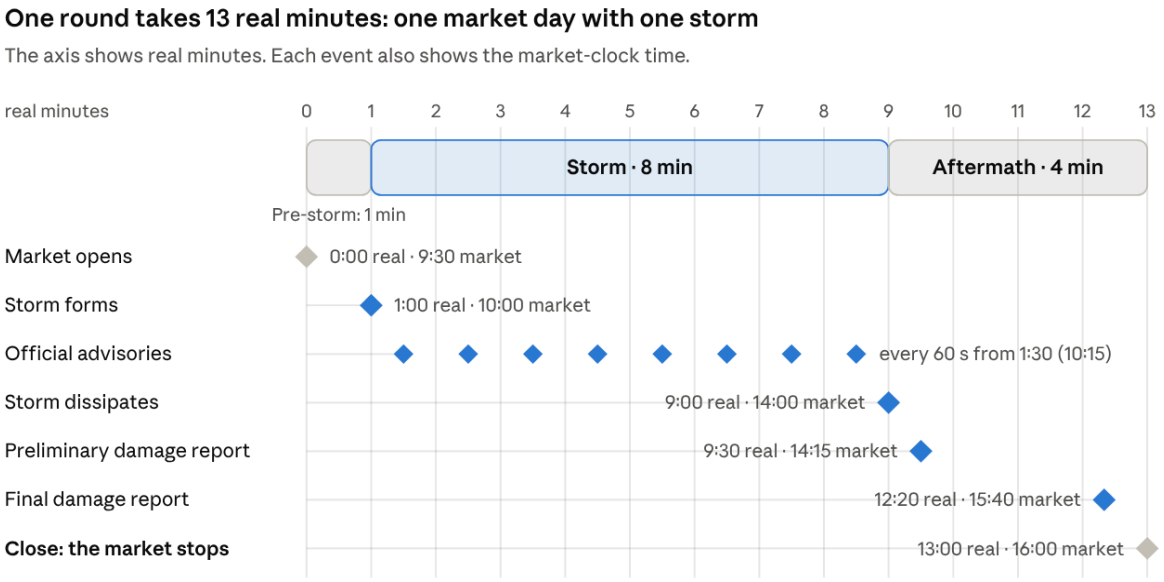


Figure 2. One round, drawn to scale. The axis is real time; each event is also labelled with its market-clock time.

3. The instruments

Twelve symbols trade: 10 stocks in three groups, and 2 ETFs. The starting price is the house price at the first open.

Ticker	Company	Group	Value depends on	Starting price	Position limit
PORT	Cascadia Pacific Terminals	Island business	1 port (Cascadia Prime)	72.48	200
ISLE	Isle Resorts International	Island business	3 resorts (Coral Key, Azure Isle, Palm Atoll)	50.26	300
AQUA	AquaVida Fisheries	Island business	2 farms, 2 fleets (Palm Atoll, Coral Key)	27.33	500
GRID	GridLock Power & Utilities	Island business	5 substations (all islands)	36.41	400
FERRY	Archipelago Ferries	Island business	4 terminals (all islands)	17.91	800
RECON	Reconex Construction	Rebuilder	Repair work	52.38	500
MAT	Materials Holdings	Rebuilder	Repair work	35.81	700
GEN	Backup Power Co.	Rebuilder	Power grid outages	40.30	600
HAVEN	HavenBridge Insurance	Insurer	Insured losses	39.55	250
REIN	Meridian Re	Insurer	Insured losses (reinsurance)	89.36	200
ARCH	Archipelago Index	ETF	1 PORT, 1 ISLE, 2 AQUA, 1 GRID, 2 FERRY, 1 HAVEN	289.20	50
REBLD	Rebuild & Recovery ETF	ETF	2 RECON, 2 MAT, 1 GEN	216.68	100

The file `case.json` gives the coordinates and the weight of each site. It also gives a description of each company. Read the descriptions closely; they contain details that affect value.

What changes the value of a company

- **Island businesses** lose value when a storm damages their sites. The damage is permanent. A closed site can also lose revenue.
- Rebuilders gain from damage to the island businesses, with diminishing returns: crews and supplies are limited, so each extra dollar of damage adds less value. Each rebuilder does a different type of work:
    - RECON repairs the port, the grid and the ferry terminals.
    - MAT supplies materials for all repairs.
    - GEN supplies power while the grid is down.

- **Insurers** pay for the insured losses of the full season. Each insurer pays only the part of the season total that is in its layer.

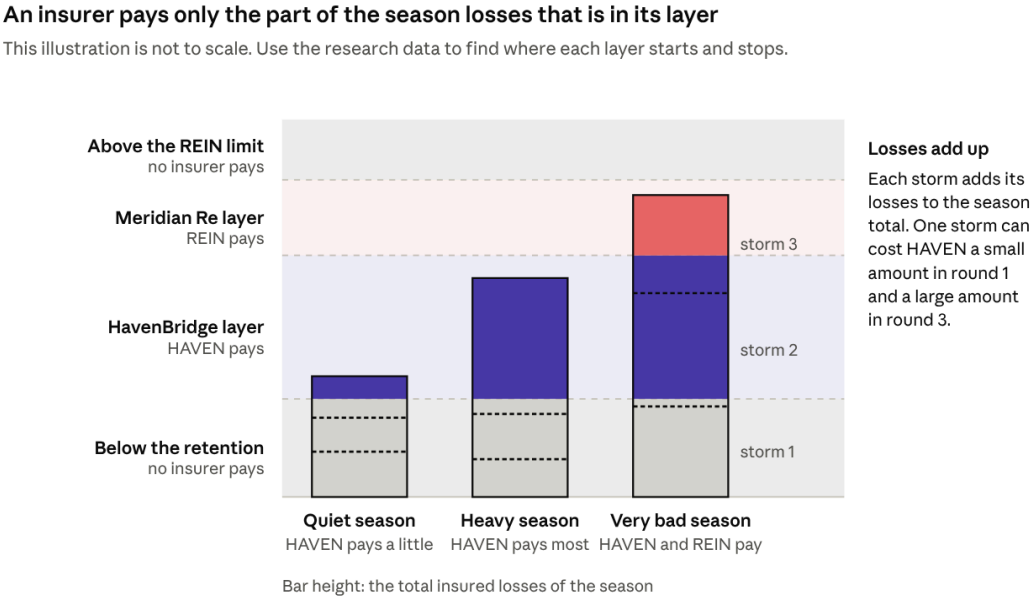


Figure 4. How insurance layers work. Not to scale; the real layer bounds are deliberately not shown.

Because of the layers (Figure 4), HAVEN and REIN can react very differently to the same storm. What an insurer pays depends on where the season total lands, not on the size of any one storm.

From the historical data, you may want to analyze:

- how easily a storm damages each site
- the cost of an outage
- the repair capacity of the rebuilders
- where each insurance layer starts and stops

ETFs

Each ETF trades on its own order book, so its price can drift away from the value of its basket.

When the market is open, you can create or redeem ETF shares:

- **Create:** give the basket and get 1 ETF share.
- **Redeem:** give 1 ETF share and get the basket.

Use the ETF panel or the API: `POST /api/v1/etf/{symbol}/create` and `POST /api/v1/etf/{symbol}/redeem`. Both operations are immediate and have no fee. The exchange books the stocks at their marks and the ETF share at the value of the basket.

You do not need to hold the basket first. A create or redeem can leave you short, as long as you stay within the position limits. Each create or redeem is one order action.

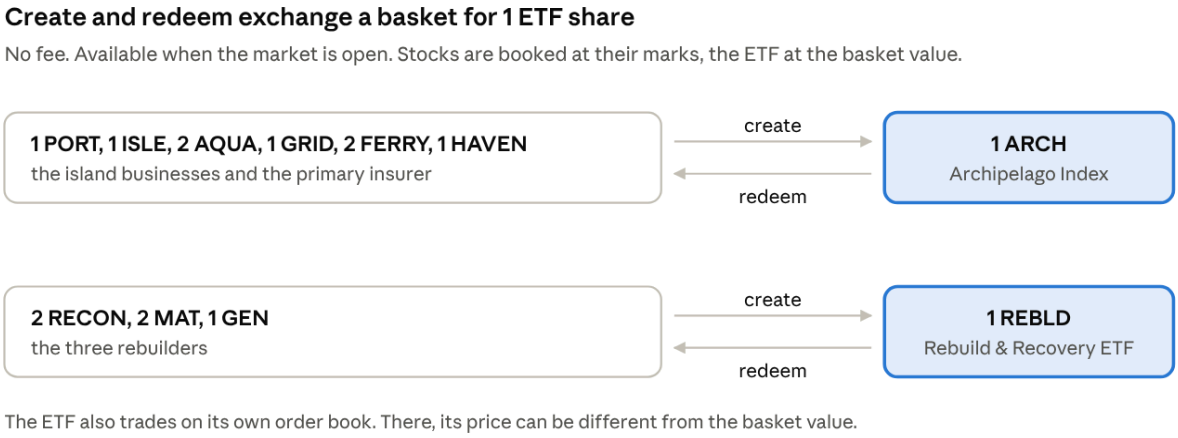


Figure 5. The two ETF baskets, as defined in `case.json`. Create and redeem have no fee and are only available while the market is open.

Settlement

At the close of round 3, after the final damage report:

- each stock settles at its fundamental value
- each ETF settles at the value of its basket

Your positions are marked at these values for scoring (Section 8).

4. What you see

All data about the current condition of a storm is public. The feed sends this data every 2 seconds. The future path of the storm is not public.

The storm

Each storm has a position, a velocity, a radius and an intensity. Wind falls off quickly with distance from the center. The radius sets how far the strong wind reaches; the intensity sets how strong it is.

A site takes damage when it is inside the wind field of a storm. Closer and stronger storms do more damage.

When the wind at a site exceeds a threshold, the site has hurricane conditions and closes. It reopens when the wind drops back below the threshold.

Storms form south of the map and usually move north. Over its last minute, a storm weakens and then dissipates.

Large-scale steering patterns move each storm. Forecasters describe them with six outlooks:

- Direct strike

- Island chain
- Recurve
- Stall
- Eyewall replacement
- Erratic

Erratic covers any storm that does not follow one of the other five patterns. Any storm can turn erratic.

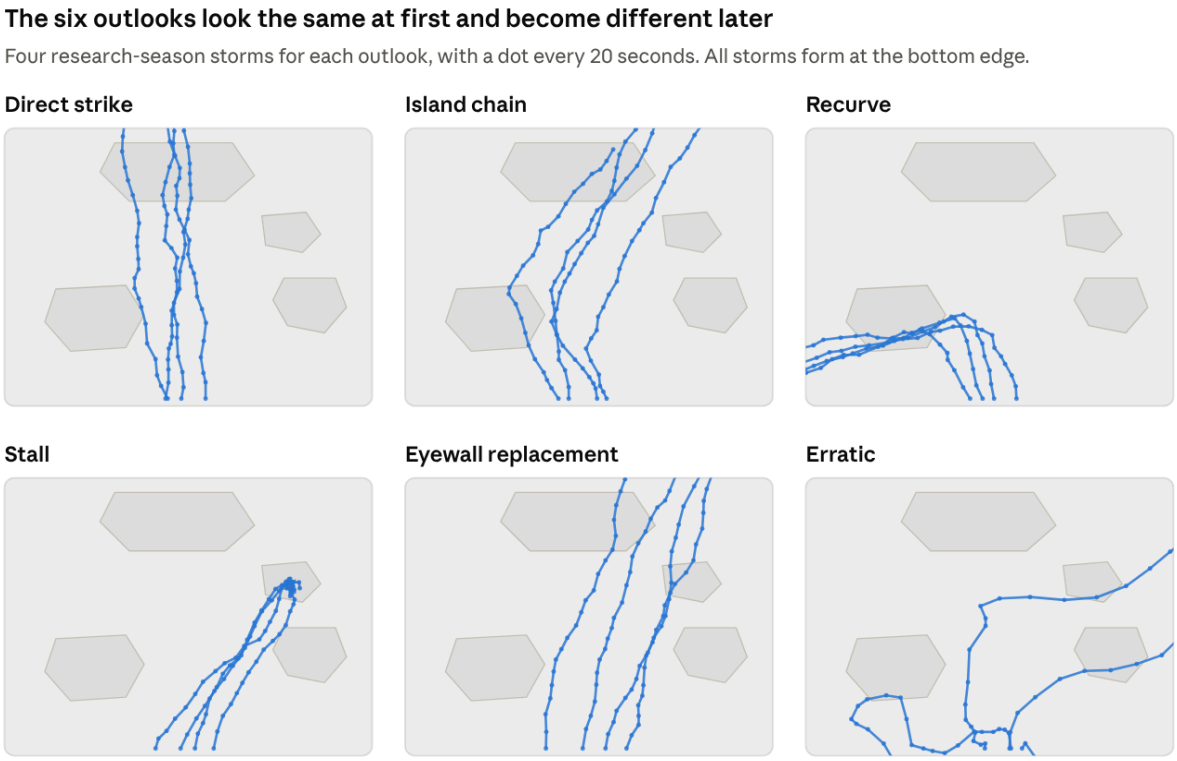


Figure 6. Sample tracks for each outlook: four storms per outlook from the research seasons, plotted with a dot every 20 seconds.

Most storms head north for their first minute (Figure 6), so Direct strike, Island chain and Recurve look alike early on. They separate only when the storm turns.

Sources

Source	Time	Contents
Storm position	Every 2 s	Position, velocity, radius and intensity
Track	Every 20 s	The path of the storm until now
Official advisory	Every 60 s, from 30 s after the storm forms	Outlook, probability of hurricane conditions on each island, forecast cone

Source	Time	Contents
Wire chatter	At random times, approximately every 45 s	A rumored outlook
Impact report	When an island first has hurricane conditions	The name of the island
Damage report	Preliminary report 30 s after the storm ends. Final report 40 s before the close.	Damage % for each company, and insured losses. The preliminary report is an estimate. The final report is exact.
News and announcements	At any time during the season	Headlines. An announcement changes the economics of a company from that time.

Storm position and track are measurements. Official advisories and wire chatter are opinions: advisories are usually right but not always, and wire chatter is less reliable. Use the historical data to measure how accurate a source is.

How you get the data

The feed comes on the market-data WebSocket, `/ws/v1/market-data`, on the news channel. Each frame contains one message. The Storm Watch screen of the trading terminal shows the same feed. Bots connect with a bot key in the `X-API-Key` header.

The static case data is at `GET /cases/hurricane.json`. It contains the map, the islands, the sites, the companies, the ETF baskets, the limits, the outlooks and the timing. The appendix documents every message type and field.

5. The market

The house provides liquidity in every symbol for the whole round, so you can always trade. It quotes around its consensus: its own estimate of each company's value. The consensus is reasonable but imperfect:

- It reacts slowly. After an announcement, it takes about 45 real seconds to adjust fully.
- Its errors show up in the research data, so you can learn them.

Three kinds of participants trade:

- **House market makers** quote bid and ask prices in each symbol around the consensus. They skew their quotes against their inventory and widen their spreads when risk rises.
- **House flow** trades for its own reasons. Some of it is noise, which picks up after news; some of it trades on wire rumors.
- **Other teams** trade on the same order books. Your orders can fill against their orders. Their trades change the prices that you see.

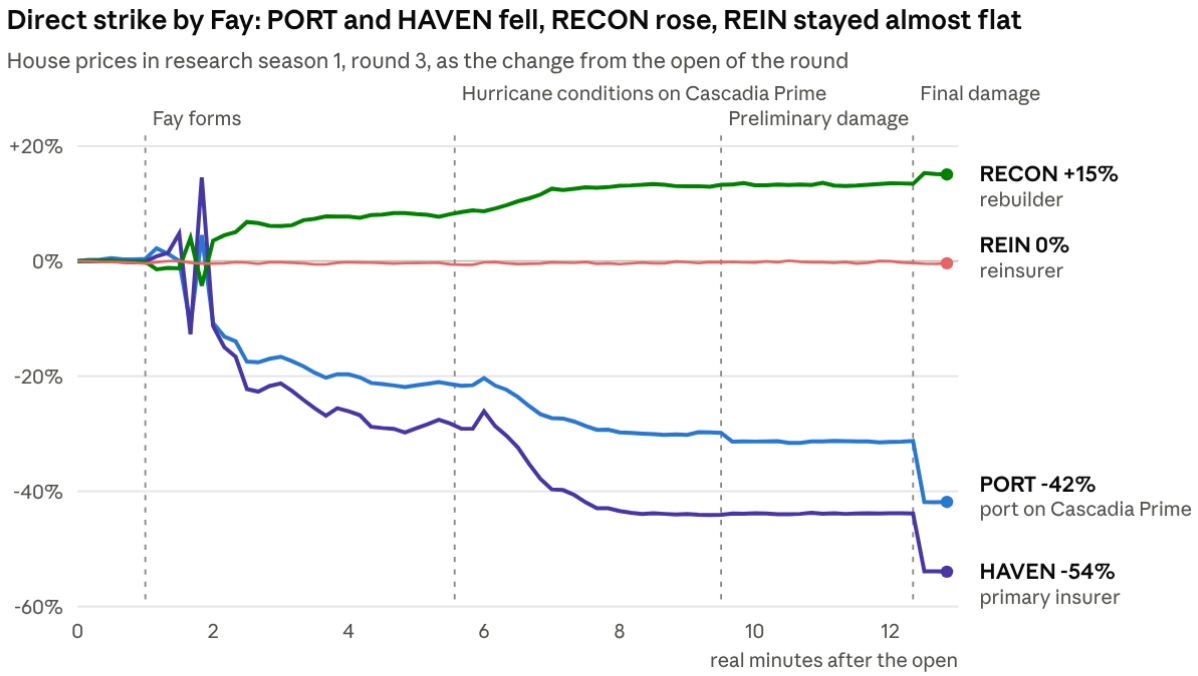


Figure 9. House prices in research season 1, round 3, as the change from the open. Dashed lines mark the storm events. Source: `market.csv`, sampled every 10 s.

Figure 9 shows round 3 of research season 1. After the first advisories for Fay, the house repriced for a direct strike within a minute. The prices changed again at the final damage report. HAVEN fell more than PORT, because HavenBridge paid most of the insured losses of the season. REIN barely moved.

6. Research data and starter kit

You get 600 past seasons, with a total of 1,800 storms. The past seasons and the live season come from the same simulator, with the same map, companies, feed and timing. Use the research data to estimate the numbers that this packet does not give:

- how damage changes the value of each company
- the accuracy of each source
- where each insurance layer starts and stops
- how the house market operates

The dates of season N are approximately N years before the live season. No storm in the research data is a storm of the live season.

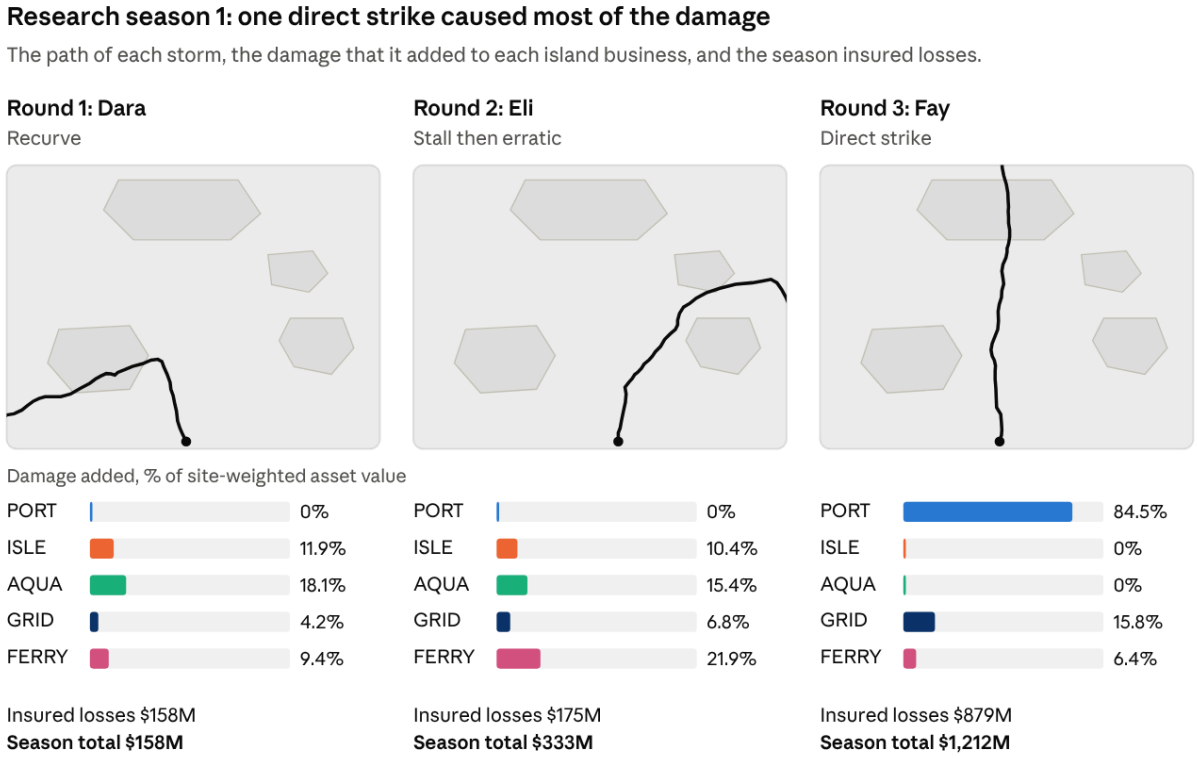


Figure 10. Research season 1, storm by storm: the path of each storm, the damage it added to each island business, and the running total of insured losses. Source: `manifest.json` and `feed.jsonl`.

Figure 10 shows research season 1. In rounds 1 and 2, storms Dara and Eli caused moderate damage on Palm Atoll, Coral Key and Azure Isle. After round 2, the insured losses of the season were $333M. In round 3, storm Fay hit the port directly and added $879M. The manifest gives the true pattern and the damage of each storm. The feed gives all the information that was available during the season.

The download, `hurricane-season-research.zip`, contains these files:

Path	Contents
`case.json`	The public case data: map, islands, sites, companies, ETF baskets, limits, outlooks and timing
`history/manifest.json`	Each season and storm, with the true pattern, the damage and the insured losses
`history/season-NNN/feed.jsonl`	The feed of the season, as the exchange published it
`history/season-NNN/market.csv`	The house prices every 2 s
`history/season-NNN/settlement.json`	The settlement value of each symbol

Path	Contents
`starter_kit/`	A data loader, a feed recorder, a recording check for the testnet and an example bot
`README.md`	The data dictionary for the files. The appendix of this packet describes the feed messages.

The research data does not include:

- News or announcements. The live season can have them.
- The model's parameters. Section 3 lists the ones you need to estimate.
- Order books. Use the testnet to see spreads, depth and the house quotes.

Reference code

The file `history.py` in the starter kit loads the data. It uses only the Python standard library. The example below measures how often the official advisories name a storm's true pattern. Run it from the `starter_kit` folder. Use it as a starting point for your own analysis.

```
from history import load

data = load("../history")
right = total = 0
for season in data.seasons:
    for storm in season.storms:
        flip = storm["turned_erratic_at"]
        for adv in season.messages(round=storm["round"], type="advisory",
source="official"):
            truth = "erratic" if flip is not None and adv["age"] >= flip else
storm["pattern"]
            right += adv["outlook"] == truth
            total += 1
print(f"official advisories named the true pattern {right / total:.0%} of the time")
```

7. The testnet

The test exchange is not ready yet, but we will update once it is ready. It is a copy of the competition exchange. It replays the research seasons, one after the other, until the competition. Use it to confirm that your bot can:

- connect to the exchange
- read each message type
- place and cancel orders
- operate correctly with the market clock

Same as the live case	Different from the live case
The exchange, its rules, order types, limits and rate limits	The storms. The testnet uses the research seasons, in sequence.
Message formats and round timing	Announcements. The research seasons have no announcements.
Tickers, map, companies and ETF baskets	Dates and results
The house market and the price protection	The pauses between rounds

Each season replays exactly as it is in the research data:

- The messages are the same, with the same `seq`.
- The house quotes around the prices of that season.
- The symbols settle at the settlement values of that season.

The market-clock date tells you which season plays. Because the replay is exact, you can verify your feed handling message by message:

1. Record the feed with `record_feed.py`.
2. Compare your recording with the research data. Use `check_recording.py`.
3. Make sure that the check prints “recording matches the published feed”.

When a season ends, the testnet starts the next one:

- Your account, team, bot keys and WebSocket connection stay the same.
- Your positions go back to zero.
- The competition gets a new id.

After each close, and after the testnet prepares a new season, the clock stops for 2 minutes. When the clock starts again, the market opens after 1 real minute.

To start on the testnet:

1. Sign in with Google.
2. Create a team (if you are the first member of your team to log in). Otherwise, join your team.
3. Ask your captain to make a bot key on the Team page.
4. Send the key in the `X-API-Key` header from your bot.
5. Send a `User-Agent` header with each HTTP request. The network in front of the exchange refuses the default `User-Agent` of the Python urllib library.

Before the event

- ☐ Download `hurricane-season-research.zip` and read `README.md`.
- ☐ Load the data with `starter_kit/history.py`. Run the reference code.
- ☐ Build a valuation model for each company. Include the effects of damage, outages and season losses.

- [ ] Measure the accuracy of the official advisories and the wire chatter. Also measure how early each one identifies the true pattern.
- [ ] Sign in on the testnet, join a team and get a bot key from your captain.
- [ ] Connect your bot. Subscribe to news, then place an order and cancel it.
- [ ] Make sure that your bot stays connected when a new season starts.
- [ ] Record a season with `record_feed.py`. Check the recording with `check_recording.py`.
- [ ] Trade a full season on the testnet, by hand and with your bot.

8. Rules and scoring

Trading rules

- **Position limits:** Each symbol has a position limit (Section 3). The limit applies to long and short positions. The limit includes your position and your resting orders on the same side. If an order goes past the limit, the exchange makes the order smaller or rejects it. If the part of an order that fills immediately needs room under the limit, the exchange cancels your resting orders on the same side, worst price first.
- **Orders:** You can send limit orders and market orders, and you can cancel orders. You cannot modify an order; cancel it and send a new one. Prices can have up to 6 decimal places. You can have a maximum of 200 resting orders in each symbol. The exchange prevents self-trades. If your new order crosses your own resting order, the exchange cancels the resting order.
- **Rate limits:**
    - 60 order actions each minute for each bot key, and 180 for each team.
    - 20 orders each minute from the browser.
    - 100 requests each second for each team, for all of the API.
    - 25 market-data connections for each team.
    - An order action is one of these: place an order, cancel an order, cancel all orders, create or redeem ETF shares.

API

All endpoints start with `/api/v1`. Send your bot key in the `X-API-Key` header.

Method and path	Use
`POST /orders`	Place an order. Body: `{"symbol": "PORT", "side": "buy", "order_type": "limit", "quantity": 10, "price": 72.5}`. A market order has no price.
`DELETE /orders/{order_id}`	Cancel one order.
`DELETE /orders/all`	Cancel all your orders. Add `?symbol=PORT` to cancel only the orders in one symbol.

Method and path	Use
`GET /orders/open`	Get your resting orders.
`GET /positions`	Get your positions, with marks and unrealized PnL.
`GET /orderbook/{symbol}`	Get the order book of a symbol.
`GET /competitions/time`	Get the market-clock time and the state of the competition.
`POST /etf/{symbol}/create` and `POST /etf/{symbol}/redeem`	Create or redeem ETF shares. Body: `{"quantity": 1}`.

The file `example_bot.py` in the starter kit shows how to use these endpoints.

Scoring

Your score is the total profit and loss (PnL) of your team after the final settlement at the end of round 3. The score includes:

- your position, marked at its settlement value
- all profit and loss that you realized during the season

The team with the highest PnL wins.

Appendix: feed reference

The feed comes on the market-data WebSocket `/ws/v1/market-data`. Connect with a bot key in the `X-API-Key` header, or with your browser session.

Send one subscription message in the first 10 seconds after you connect, for example `{"action": "subscribe", "symbols": [], "channels": ["news"]}`. To get market data, add symbols and the `quotes`, `trades` or `orderbook` channels to the same message. The exchange ignores all subsequent messages. If a client sends more than 10 frames in one second, the exchange closes the connection.

Each frame contains one message, in the form `{"type": "news", "message": {...}}`. The feed does not replay old messages. A client that connects late gets only the messages that the exchange publishes after it subscribes. A snapshot comes every 30 s. The examples below are from research season 1.

Each message has these fields:

Field	Meaning
`type`	`storm`, `track`, `storm_formed`, `storm_dissipated`, `impact`, `advisory`, `damage_report`, `news`, `announcement` or `snapshot`
`round`	The round, 1 to 3
`storm`	The name of the storm of that round
`time`	The market-clock time (fictional US/Eastern)
`seq`	The position of the message in the feed of the season. The value increases. A snapshot has no `seq`. Gaps are normal. Ignore a message if you already have its `seq`.
`headline`	Text for people. All types have a headline, except storm, track and snapshot.

Map coordinates: the map is 1,000 by 700 units. The origin is at the top left, and y increases to the south. Storms form south of the map, at y = 740, and move north. The age of a storm is in real seconds since it formed.

`storm`: one observation of the live storm. `radius` is the size of the storm. `intensity` is its strength, which decreases in the last minute of the storm. `vx` and `vy` give the velocity in map units each second.

```
{"type": "storm", "round": 1, "storm": "Dara", "time": "2025-08-04T10:00:00", "age": 0,
"x": 480.9, "y": 740.0, "vx": 0.0, "vy": -1.8, "radius": 80.0, "intensity": 1.0, "seq":
1}
```

`track`: the full path of the storm until now, as `[age, x, y, radius]` points every 4 seconds of storm age. Use the track to fill gaps in the observations.

```
{"type": "track", "round": 1, "storm": "Dara", "time": "2025-08-04T10:10:00", "age":
20, "points": [[0, 480.9, 740.0, 80.0], [4, 478.3, 732.7, 79.7], [8, 473.9, 725.8,
78.2], "..."], "seq": 12}
```

`storm_formed` and `storm_dissipated`: the start and the end of the storm.

```
{"type": "storm_formed", "round": 1, "storm": "Dara", "time": "2025-08-04T10:00:00",
"headline": "Dara has formed south of the archipelago.", "seq": 0}
```

`impact`: the first hurricane conditions on an island. The island keys are the same as in `case.json`. Each storm sends a maximum of one impact message for each island.

```
{"type": "impact", "round": 1, "storm": "Dara", "time": "2025-08-04T11:02:00",
"island": "palm_atoll", "age": 124, "headline": "Hurricane conditions reported on Palm
Atoll.", "seq": 76}
```

`advisory` with `source: "official"`: the numbered advisory of the forecast office.

- **`outlook`:** the steering pattern that the forecast office expects. The `outlooks` list in `case.json` gives the keys and names.
- **`hurricane_conditions`:** the probability of hurricane conditions on each island during the remainder of the storm.
- **`cone`:** the forecast track. Each point gives the expected position at `age` seconds into the storm. `spread` is the typical distance of the storm from that point.

```
{"type": "advisory", "round": 1, "storm": "Dara", "time": "2025-08-04T10:15:00",
"source": "official", "number": 1, "age": 30, "outlook": "chain", "outlook_name":
"Island chain", "hurricane_conditions": {"cascadia_prime": 0.78, "coral_key": 0.02,
"azure_isle": 0.0, "palm_atoll": 0.8}, "cone": [{"age": 60, "x": 441.1, "y": 628.6,
"spread": 13.6}, {"age": 90, "x": 419.3, "y": 577.5, "spread": 22.1}, "..."],
"headline": "Advisory 1: steering outlook Island chain. Hurricane conditions: Palm
Atoll 80%, Cascadia Prime 78%.", "seq": 17}
```

`advisory` with `source: "wire"`: a rumor about the outlook of the storm. It uses the same outlook keys. It has no probabilities and no cone.

```
{"type": "advisory", "round": 1, "storm": "Dara", "time": "2025-08-04T10:17:00",
"source": "wire", "age": 34, "outlook": "recurve", "headline": "Wire: forecasters
chatter that Dara will recurve out to sea.", "seq": 21}
```

`damage_report`: the damage that this storm added to each island business, as a percentage of its site-weighted asset value. It also gives the insured losses of this storm in $M. The preliminary report is an early estimate. The final report is the assessed value.

```
{"type": "damage_report", "round": 1, "storm": "Dara", "time": "2025-08-04T14:15:00",
"stage": "preliminary", "damage_pct": {"PORT": 0.0, "ISLE": 9.5, "AQUA": 21.2, "GRID":
6.4, "FERRY": 9.7}, "industry_loss_musd": 166.3, "headline": "Preliminary damage
estimates for Dara: insured losses near $166M.", "seq": 289}
{"type": "damage_report", "round": 1, "storm": "Dara", "time": "2025-08-04T15:40:00",
"stage": "final", "damage_pct": {"PORT": 0.0, "ISLE": 11.92, "AQUA": 18.08, "GRID":
4.23, "FERRY": 9.37}, "industry_loss_musd": 157.5, "headline": "Final damage assessment
for Dara: insured losses $158M.", "seq": 290}
```

`news`: a headline. **`announcement`:** a company or government action that changes the economics of a company from that time. `symbols` lists the companies. The research seasons have no news and no announcements. An `announcement` has this shape:

```
{"type": "announcement", "round": "<round>", "storm": "<storm>", "time": "<market
time>", "headline": "<what changed>", "symbols": ["<ticker>"], "seq": 0}
```

`snapshot`: the current condition of the round. The feed sends a `snapshot` every 30 s, at the open of each round and after a catch-up. A screen or a bot that connects late uses the `snapshot` to get the current condition. The `snapshot` describes the current round, or the round that closed last.

Field	Meaning
`time`	The market-clock time of the snapshot
`phase`	`pre_storm`, `storm`, `aftermath` or `closed`
`last_seq`	The snapshot includes all feed messages up to this `seq`
`obs`	The latest storm message (without type, round and storm), or `null`
`track`	Approximately 40 track points `[age, x, y, radius]` that end at `obs`
`advisory`	The latest official advisory of the round, or `null`
`impacts`	`[{island, age, time}]` for this round
`damage`	All damage reports until now, from all rounds, oldest first
`headlines`	The last 8 headline messages, from all rounds

Delivery: the messages arrive in `seq` order. After a problem on the exchange, the missed messages arrive together and in order. For `storm` and `track`, only the latest message of the group arrives. After a long gap, a snapshot follows. Rarely, a message arrives two times after a network retry. Use `seq` to remove duplicates.